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Randomized Placebo-Controlled Trial of Nabilone for Agitation in Alzheimer's Disease

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ABSTRACT

Objective: *To investigate the efficacy and safety of nabilone for agitation in patients with moderate-to-severe Alzheimer's disease (AD).* ***Design:*** *This 14-week randomized double-blind crossover trial compared nabilone to placebo (6 weeks each) with a 1-week washout between phases.* ***Setting:*** *Patients were recruited from a long-term care facility and geriatric psychiatry clinics.* ***Participants:*** *Patients had AD (standardized Mini-Mental State Examination [sMMSE ≤24]) and agitation (Neuropsychiatric Inventory-Nursing Home version [NPI-NH]-agitation/aggression subscore 3).* ***Intervention:*** *Nabilone (target 1–2 mg) versus placebo.* ***Measurements:*** *The primary outcome was agitation (Cohen Mansfield Agitation Inventory [CMAI]). Secondary outcomes included NPI-NH total, NPI-NH caregiver distress, cognition (sMMSE and Severe Impairment Battery [SIB] or Alzheimer's Disease Assessment Scale of Cognition), global impression (Clinician's Global Impression of Change [CGIC]), and adverse events.* ***Results:*** *Thirty-nine patients (mean ± SD age = 87 ± 10, sMMSE = 6.5 ± 6.8, CMAI = 67.9 ± 17.6, NPI-NH total = 34.3 ± 15.8, 77% male, nabilone dose = 1.6 ± 0.5 mg) were randomized. There were no crossover or treatment-order effects. Using a linear mixed model, treatment differences (95% CI) in CMAI (b = −4.0 [−6.5 to −1.5], t(30.2) = −3.3, p = 0.003), NPI-NH total (b = −4.6 [−7.5 to −1.6], t(32.9) = −3.1, p = 0.004), NPI-NH caregiver distress (b = −1.7 [−3.4 to −0.07, t(33.7) = −2.1, p = 0.041), and sMMSE (b = 1.1 [0.1–2.0], t(22.6) = 2.4, p = 0.026) all favored nabilone. However, in those who completed the SIB (n = 25) treatment differences*

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favored placebo (b = −4.6 [−7.3 to −1.8], t(20.7) = −4.8, p = 0.003). CGIC improvement during nabilone (47%) and placebo (23%) was not significantly different (McNemar's test, exact p = 0.09). There was more sedation during nabilone (45%) compared to placebo (16%) phases (McNemar's test, exact p = 0.02), but treatment-limiting sedation was not significantly different (McNemar's test, exact p = 0.22). ***Conclusions:*** *Nabilone may be an effective treatment for agitation. However, sedation and cognition should be closely monitored.* (Am J Geriatr Psychiatry 2019; 27:1161−1173)

INTRODUCTION

Agitation, seen in 20%−50% of those with moderate-to-severe Alzheimer's disease (AD),[1,2] is particularly common in long-term care facility residents[3,4] and a challenging neuropsychiatric symptom (NPS) to treat. Agitation is associated with decreased quality of life, increased caregiver burden, and higher rates of institutionalization and mortality.[3,5] Nonpharmacologic interventions are considered first-line therapy for the management of agitation in AD. However, with severe agitation, when nonresponsive to nonpharmacologic interventions, judicious use of pharmacologic interventions is recommended.[3] These include psychotropics[3] such as atypical antipsychotics, which have consistent but modest efficacy for agitation, and are associated with increased risk of cerebrovascular adverse events and mortality.[6] There is a pressing need to identify safer and more efficacious pharmacotherapies for agitation in AD.

Cannabinoids (CBs) such as tetrahydrocannabinol (THC) are agonists at cannabinoid receptors 1 and 2 (CB1/2), and have psychotropic effects that may benefit AD patients with agitation.[7−10] However, THC use has been associated with cognitive deficits and psychosis in non-AD populations[11] and lack of efficacy in AD.[12,13] The synthetic oral THC analogue nabilone, as a partial agonist at CB1/2, may have therapeutic potential for the treatment of agitation in patients with moderate-to-severe AD.

The primary aim of this study was to evaluate the efficacy and safety of nabilone for agitation in patients with moderate-to-severe AD. Other aims included investigating the efficacy of nabilone for overall NPS, cognition, clinical global impression, nutritional status, pain, and caregiver burden.

METHODS

Primary Research Question/Classification of Evidence

This double-blind randomized controlled trial (RCT) investigated the efficacy and safety of nabilone compared to placebo for agitation in patients with moderate-to-severe AD. This study provides Class I evidence of the efficacy and safety of nabilone compared to placebo for treating agitation in patients with moderate-to-severe AD.

Standard Protocol, Approvals, Registrations, and Participant Consents

Approval was obtained from the institutional Research Ethics Board (REB) and Health Canada. This study was registered at ClinicalTrials.gov (NCT02351882). Study procedures were conducted in accordance with the Declaration of Helsinki and the International Conference on Harmonisation Consolidated Guidelines on Good Clinical Practice. Informed consent was obtained following procedures outlined by the institutional REB, which are consistent with those of the Alzheimer's Association.[14] Specifically, given the advanced cognitive impairment status of the enrolled study participants, study participation was approved by the patient's substitute decision maker and by the patient's treating physician.

Study Participants

Study participants were recruited from the Sunnybrook Health Sciences Centre long-term care facility and outpatient geriatric psychiatry clinics across the

Greater Toronto Area. Eligible participants were 55 years old or older, met the *Diagnostic and Statistical Manual of Mental Disorders* (DSM)-5 criteria for Major Neurocognitive Disorder due to AD, or met both Major Neurocognitive Disorder due to AD and Major Vascular Neurocognitive Disorder. Patients also had a score of less than or equal to 24 on the standardized Mini-Mental Status Examination (sMMSE) and clinically significant agitation/aggression (Neuropsychiatric Inventory (NPI)-agitation/aggression subscore ≥3). Patients had to be on a stable dose of a cognitive enhancer (cholinesterase inhibitors or memantine) for greater than or equal to 3 months and psychotropics for greater than 1 month prior to randomization if used. Patients were excluded if they had contraindications to nabilone (history of hypersensitivity to any cannabinoid, and/or clinically significant and unstable cardiovascular disease), or the presence or history of other neurologic conditions or other psychiatric disorders, including clinically significant delusions and/or hallucinations (NPI-delusions or hallucinations subscore ≥4).

Study Procedures

This was a double-blind, 14-week crossover trial comparing 6 weeks of nabilone (therapeutic dose: 1−2 mg per day) and matching placebo with a 1-week single-blind placebo phase preceding each treatment phase. In the treatment phase, after the placebo run-in, participants received 0.25 mg before bedtime for three nights, then titrated up to 0.5 mg/day total for the next 4 days, and maintained this dose for week 1. In week 2, participants were titrated up to 1 mg/day. During weeks 3 and 4, participants received a flexible dose of 1−2 mg/day based on tolerability. Tolerability was assessed by the study physician based on clinical assessments and caregiver reports. The maximum target dose was 2 mg/day. In the event of an intolerable adverse event potentially related to the study medication, the study physician decreased the dose by 0.5 mg increments until resolution. In the event that side effects were still intolerable at 1 mg/day, the study medication was discontinued. The tolerated dose was maintained until down-titration in week 6. Following 3 days of maintenance dose during week 6, participants were tapered down for the remaining 4 days to reduce the risk of potential withdrawal. On days 4 and 5 of week 6, participants received 1 mg/day, and on days 6 and 7, participants received 0.5 mg/day total. The placebo phase followed the same dosing structure. Early crossover was permitted for tolerability or efficacy issues.

A block randomization code was independently computer generated by the institutional Pharmacy Department. Patients, family members, nurses, clinicians, outcome assessors, and investigators were blinded to treatment allocation and block size. Study staff were unblinded only when the final patient completed all study assessments and the database was locked. Unblinding during the study was permitted only in exceptional clinical circumstances (i.e., necessary for acute medical management of serious adverse events [SAE]) and only after approval by the principal or qualified investigator (PI or QI).

Eligible and consented patients were assessed on the following weeks: −1 (screening; placebo run-in), 0 (baseline, start of treatment 1), 2, 4, 6 (end of treatment 1; start of placebo washout), 8 (start of treatment 2), 10, 12, 14 (end of treatment 2; final assessment), and 15 (safety follow-up). The Cohen-Mansfield Agitation Inventory (CMAI), NPI-Nursing Home (NPI-NH), Mini-Nutritional Assessment Short-Form (MNA-SF), and Pain Assessment in Alzheimer's disease (PAIN-AD) were administered by trained study staff to family (outpatients) or primary/associate nurses (inpatients) at every study visit. Cognition was assessed at the start and end of each treatment phase (weeks 0, 6, 8, and 14) using the sMMSE, and either the Severe Impairment Battery (if sMMSE ≤15) or the Alzheimer's Disease Assessment Scale of Cognition (ADAS-Cog) (if sMMSE >15). These assessments were administered by trained study staff to study participants. The study physician completed the Clinician's Global Impression (CGI) at each baseline, and the Clinician's Global Impression of Change (CGIC) at subsequent visits. Vital signs (blood pressure, pulse rate, and weight) were obtained at each study visit to monitor safety.

Outcomes Measures

The primary efficacy outcome measure of agitation was the CMAI, a 29-item scale that measures 4 domains of agitation: physical aggression (Questions 1−11), physical nonaggression (Questions 12−21), verbal aggression (Questions 22−24), and verbal nonaggression (Questions 25−29).[15] Each item is rated on

a scale of frequency from 1 (never) to 7 (several times an hour). Higher scores indicate greater agitation severity.

Secondary outcome measures included cognition, NPS, and global change. Cognition was assessed using the sMMSE.[16] Out of 30, lower scores indicate greater impairment. Compared to the widely used MMSE, the sMMSE imposes strict guidelines for administration and scoring to improve its inter-rater and intrarater reliability.[16] Additionally, compared to the MMSE, the sMMSE contains three different words for the registration task ("ball, car, man" versus "apple, table, penny") in order to have alternate forms of the three words, and utilizes the task in which the patient is asked to spell "world" backward as the preferred concentration task over serial sevens, providing a comprehensive chart for easy scoring. In patients with sMMSE less than or equal to 15, cognition was also assessed using the Severe Impairment Battery (SIB).[17] Similar to the sMMSE, lower scores mean greater impairment. The ADAS-Cog was completed in patients with sMMSE greater than 15.[18] Higher scores indicate greater impairment. NPS and caregiver burden were assessed with the NPI-NH, which includes an assessment of the following behavioral disturbances: delusions, hallucinations, apathy, depression, agitation, euphoria, aberrant motor behavior, irritability, disinhibition, anxiety, sleeping, and eating.[19] The greater the NPI-NH score, the more severe and frequent the behavioral disturbances. Caregiver burden was assessed with the NPI-NH caregiver distress score, with greater scores reflecting greater burden. In addition, we assessed whether patients met International Psychogeriatric Association (IPA) Provisional Research Criteria for Agitation,[20] which was defined as either physical aggressive, verbal aggressive, or physical nonaggressive symptoms that cause significant emotional distress and are frequently occurring for at least 2 weeks. We also evaluated the corresponding behaviors by taking the sum of three CMAI subscales (CMAI IPA score): physical aggression, verbal aggression, and physical nonaggression. Global impression was assessed using the CGI and CGIC, a clinician's rated seven-point Likert scale of global deterioration (1 = normal, not at all ill, 2 = borderline ill, 3 = mildly ill, 4 = moderately ill, 5 = markedly ill, 6 = severely ill, and 7 = among the most extremely ill). This scale requires the clinician to consider changes in the cognitive, behavioral, and/or functional status of the patient.[21]

Exploratory outcomes included nutritional status and pain. Nutritional status was assessed using the MNA-SF, BMI, and weight (kg). The MNA-SF categorizes patients as malnourished, at risk of malnutrition, or normal nutritional status based on 6 questions that take into account recent changes in appetite, weight loss, mobility, psychological stress/acute disease, current neuropsychological problems, and BMI.[22] Pain was assessed with the PAINAD, which assesses breathing, vocalizations, facial expression, body language, and consolability.[23]

Safety was determined by the type and number of individuals who experienced treatment emergent adverse events (TEAEs), and the total TEAEs experienced. All emergent AEs were noted and followed up until resolution.

Sample Size

A sample size calculation was performed for the primary efficacy outcome variable using a paired t test. To detect medium-to-large effect sizes (Cohen's d $\geq$0.5) for CMAI change scores between nabilone and placebo, a total of 30 patients would be needed for an alpha of 0.05 and power of 0.80. In an earlier placebo-controlled crossover trial,[24] improvements in CMAI scores favored cannabinoid treatment in 15 patients. To account for possible attrition rates (5%,[25] 14%,[26] and 27%[27] in our previous trials in similar patient populations), we aimed to randomize 40 subjects (10 drop-outs = 25% attrition).

Statistical Analysis

All analyses were performed on a modified intention-to-treat population using data from participants who took at least one dose of study medication after randomization. Paired data from all participants were used without imputation. For the primary outcome, differences between treatment phases on CMAI scores were determined using a linear mixed model with participants as a random factor and block (8 levels: weeks 0 (BL), 2, 4, 6, 8, 10, 12, and 14) and treatment (2 levels: nabilone and placebo) as fixed factors. A Cohen's d effect size was calculated using the methodology outlined by Cohen 1998 to determine whether the effect observed was small (Cohen's

d = 0.20), medium (Cohen's d = 0.5), or large (Cohen's d = 0.80).[28] Carry-over effects were determined using a paired t test between CMAI total scores at the baselines of phase 1 (week 0) and phase 2 (week 8). Treatment-order effects were determined using a t test between change in CMAI scores between nabilone in phase 1 data versus nabilone in phase 2 data.

Secondary and exploratory outcomes of efficacy on overall NPS, NPI-agitation/aggression subscore, caregiver distress, CMAI IPA subdomain scores, cognition, nutritional status, and pain were analyzed similarly to the primary efficacy analysis. For the secondary efficacy outcome of global change, CGIC ratings at the ends of each treatment phase were dichotomized to any improvement and no change/any deterioration, and between-group treatment differences were determined using McNemar's test.

To assess the safety of nabilone, the number of participants who experienced a TEAE, SAE, sedation/lethargy, and treatment-limiting sedation (e.g., sedation that led to discontinuation during a treatment phase), were compared between treatment groups using McNemar's test. In the subgroup of individuals who demonstrated improved sedation following a dose reduction, a linear mixed model was run to determine treatment differences on CMAI scores. Treatment-limiting sedation was also added as a covariate with treatment group in a linear mixed model with CMAI score as the dependent variable. The effect of treatment, treatment-limiting sedation, and the treatment x treatment-limiting sedation interaction term was computed.

Post-hoc analyses were conducted to determine whether nabilone dose was associated with differences between treatment phases on CMAI scores. Nabilone dose (1/1.5 mg versus 2 mg) was added as a covariate with treatment group in a linear mixed model with CMAI score as the dependent variable, and the effect of treatment, nabilone dose, and the treatment x nabilone dose interaction term was computed. We investigated whether concomitant psychotropic use, specifically, use of atypical antipsychotics, antidepressants, selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), or other antidepressant (trazodone or mirtazapine) was associated with differences between treatment phases on CMAI scores. Specifically, psychotropic use was added as a covariate with treatment group in a linear mixed model with CMAI score as the dependent variable. The linear mixed model computed the effect of treatment, psychotropic use, and the treatment x psychotropic use interaction term.

Statistical analyses were performed using IBM SPSS Statistics Version 20. All p-values were 2-sided, and p less than 0.05 was the threshold for statistical significance.

Data Availability Statement

Anonymized grouped data will be shared upon request by any qualified investigator.

RESULTS

Study Patient Characteristics

Thirty-nine institutionalized patients with moderate-to-severe AD were enrolled between September 2015 and December 2017. One participant discontinued during the placebo run-in (week 1) due to clinically significant delusions, and therefore 38 participants were included in the analyses. Of those, two died prior to study completion, and nine were terminated early due to a SAE (five during the nabilone phase, four during placebo; Fig. 1). The mean tolerated dose of nabilone was 1.6 ± 0.5 mg (53% 2 mg/day; 13% 1.5 mg/day; and 34% 1 mg/day). Table 1 provides baseline demographics and clinical and behavioral characteristics. All but one participant met IPA research diagnostic criteria for agitation.[20]

Study Outcomes

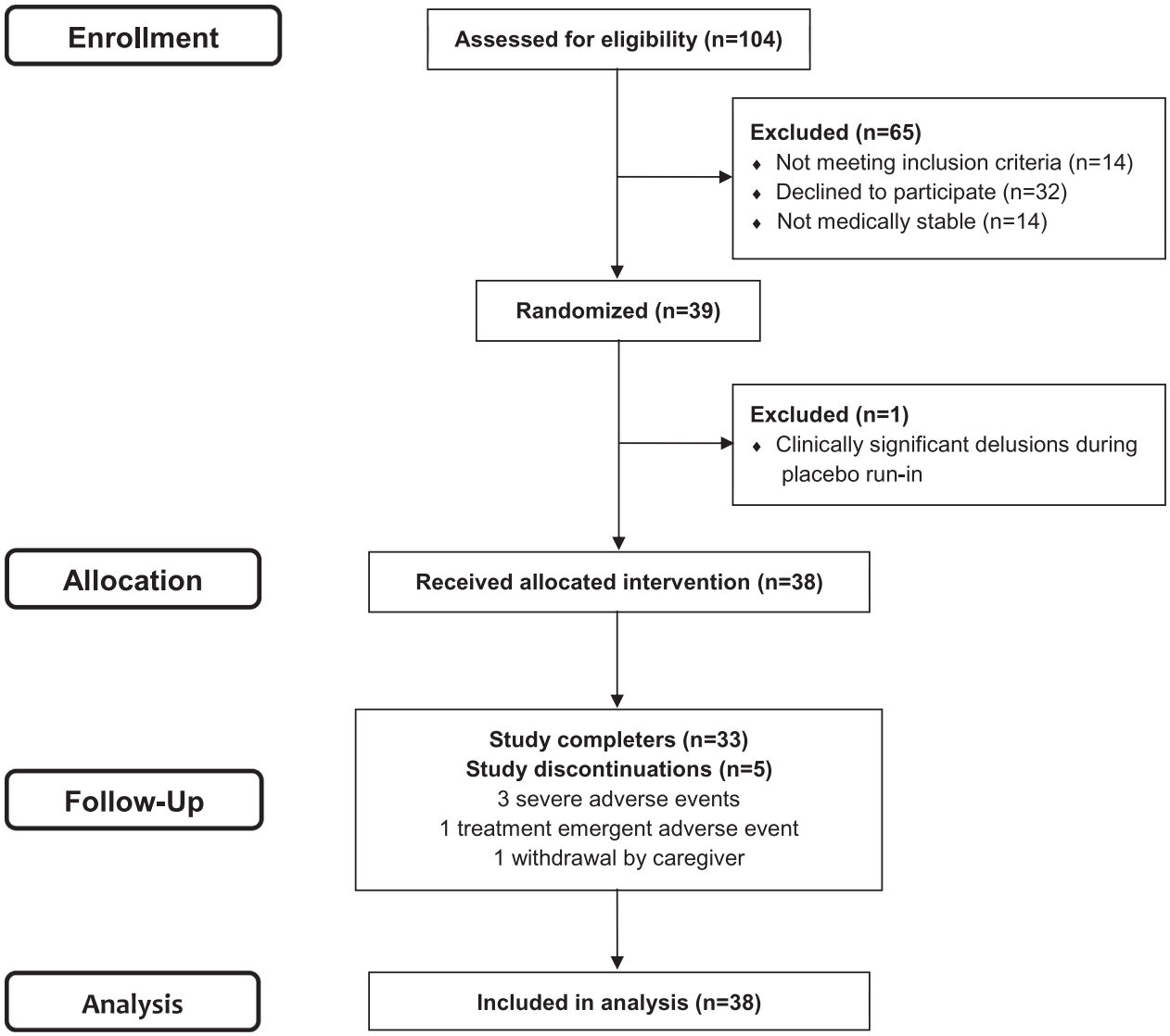
Primary (Agitation)

The estimated treatment difference (95% CIs) in CMAI scores was b = −4.0 (−6.5 to −1.5], t(30.2) = −3.3, p = 0.003 favoring nabilone over placebo (negative differences favor nabilone). This translates to a Cohen's d of 0.52, which represents a medium effect size. There were no carry-over (t(32) = 1.6, p = 0.11) or treatment order (t(31) = 0.2, p = 0.85) effects (Fig. 2).

Secondary (Behavior, Cognition, and Global Status)

There were significant differences in overall NPS as assessed by the NPI-NH (b = −4.6 [−7.5 to −1.6], t (32.9) = −3.1 p = 0.004), and in NPI-agitation/aggression

FIGURE 1. Patient flow Diagram.



subscore (b = −1.5 [−2.3 to −0.62], t(33.2) = −3.6, p = 0.001), both of which favored nabilone. NPI-NH caregiver distress scores were significantly different between treatment phases (b = −1.7 [−3.4 to =0.7], t(32.9) = −3.1, p = 0.041), favoring nabilone. The estimated treatment difference in CMAI IPA subdomain scores (physical aggression + physical nonaggression + verbal aggression) was also significant, favoring nabilone over placebo (b = −3.8 [−5.8 to −1.7], t (28.9) = −3.8, p = 0.001). There was also a significant difference in cognition (sMMSE) (b = 1.1 [0.1−2.0], t(22.6) = 2.4, p = 0.026) which favored nabilone. On the CGIC, 47% of patients demonstrated "minimal" to "marked" improvement during the nabilone treatment phase, whereas 23% of patients experienced improvement during the placebo phase (McNemar's test, exact p = 0.09) (Fig. 3).

In the subgroup with sMMSE less than or equal to 15 (n = 25), there was a significant difference in SIB score (b = −4.6 [−7.3 to −1.8], t(13.2) = −3.6, p = 0.003) that favored placebo. ADAS-Cog scores were not analyzed as only three patients completed this scale, and only one participant had scores for both treatment phases.

TABLE 1. Baseline Characteristics (N = 38)

Baseline Demographics	
Age	87 ± 10
Sex (%M)	77%
In/outpatient (% inpatient)	72%
Number of concomitant illnesses	6 ± 3
Number of concomitant medications	12 ± 5
Number of concomitant psychotropic medications	1.8 ± 0.7
Cholinesterase inhibitor	53%
Memantine	29%
Atypical antipsychotic	45%
Antidepressant	87%
SSRI	42%
SNRI	5%
Mirtazapine or trazodone	45%
Benzodiazepine	5%
Baseline characteristics	
CMAI	67.9 ± 17.6
Met IPA criteria for agitation/aggression	97%
NPI-NH total	34.3 ± 15.8
NPI-NH agitation/aggression	7.1 ± 3.3
NPI-NH total caregiver distress score	12.7 ± 7.9
sMMSE	6.5 ± 6.8
SIB (sMMSE ≤15) (n = 28)	36.5 ± 30
ADAS-Cog (sMMSE >15) (n = 3)	22.7 ± 3.1
MNA-SF	8.7 ± 2.9
PAINAD	2.6 ± 1.4
CGI severity	3.7 ± 0.9
Moderately ill	50%
Markedly ill	29%
Severely ill	18%
Extremely ill	3%
Weight (kg)	67.9 ± 14.1
BMI	24.5 ± 3.9

All numbers indicate mean and SD or percentage as indicated. SSRI: selective serotonin, selective serotonin reuptake inhibitor; SNRI: serotonin and norepinephrine reuptake inhibitors; CMAI: Cohen Mansfield Agitation Inventory; IPA: International Psychogeriatric Association; NPI-NH: Neuropsychiatric Inventory-Nursing Home version; sMMSE: standardized Mini-Mental Status Examination; SIB: Severe Impairment Battery; ADAS-Cog: Alzheimer's Disease Assessment Scale of Cognition; MNA-SF: Mini-Nutritional Assessment-Short Form; PAINAD: Pain Assessment in Alzheimer's disease; CGI: Clinician's Global Impression; BMI: body mass index.

Exploratory (Nutritional Status and Pain)

There were significant treatment phase differences on the MNA-SF (b = 0.2 [0.02−0.4], t(31.1) = 1.3, p = 0.03), favoring nabilone. However, there were no differences in weight (b = 0.01 [−0.69 to 0.71], t (30.8) = 0.03, p = 0.97) or BMI (b = −0.14 [−0.35 to 0.07], t(31.6) = −1.5, p >0.05). There were no treatment differences on the PAINAD scale (b = 0.03 [−0.22 to 0.27], t(19.9) = 0.2, p = 0.82).

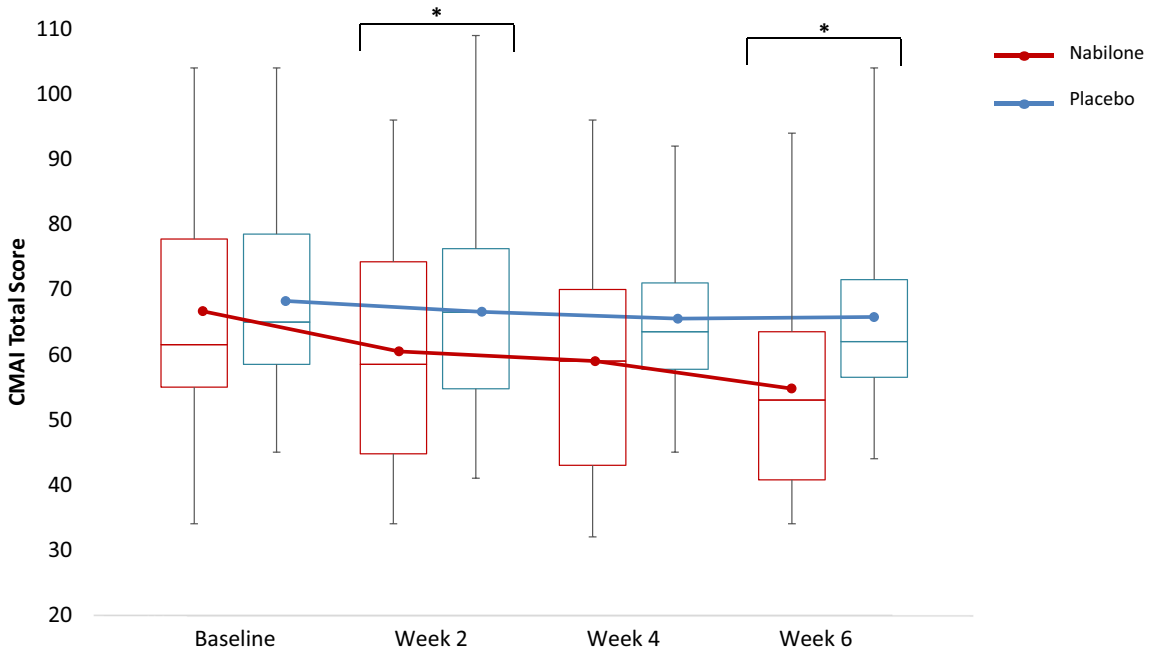
Safety

The occurrence of TEAEs was significantly greater during the nabilone phase (McNemar's test, exact p = 0.05), with sedation being the most common. More patients experienced sedation during nabilone (N = 17) compared to placebo (N = 6) (McNemar's test, exact p = 0.02). However, of the 17 in nabilone, sedation improved in 12 following a reduction in nabilone dose (e.g.,: 2−1.5 mg, or 1.5−1 mg). In those 12 patients, there was a significant treatment difference in CMAI score (b = −6.31 [−8.51 to 4.12], t (7.4) = −6.7, p <0.001), favoring nabilone. There were no significant differences in the occurrence of treatment-limiting sedation in nabilone compared to placebo (McNemar's test, exact p = 0.22). When treatment-limiting sedation was added as a covariate with treatment group in a linear mixed model with CMAI score as the dependent variable, there was a significant treatment difference in CMAI score (b = −5.0 [−8.69 to −1.32], t(28.3) = −2.8, p = 0.009) favoring nabilone, and neither treatment-limiting sedation nor the treatment x treatment-limiting sedation interaction term significantly contributed to the model. There were no significant differences in the occurrence of falls (McNemar's test, exact p = 1.0) or SAEs (McNemar's test, exact p = 0.69) between treatment groups. There was one death each in the nabilone and placebo phase, which the QI rated as possibly related to study drug. Of these, 1 death occurred during the placebo washout phase (sudden death) following 6 weeks of the placebo phase, and one participant died on the last day of nabilone treatment (suspected stroke), after being titrated down to 0.5 mg. Table 2 lists the type and frequency of TEAEs and SAEs in the nabilone and placebo phases.

Post-Hoc Analyses

When nabilone dose was added as a covariate with treatment group in a linear mixed model with CMAI score as the dependent variable, there was a significant treatment difference in CMAI scores (b = −3.8 [−7.47 to −0.13], t(25.7) = −2.1, p = 0.04) favoring nabilone, and neither nabilone dose nor the treatment x nabilone dose interaction term significantly contributed to the model. When use of atypical antipsychotics, antidepressants, SSRIs, SNRIs, and other antidepressants (trazodone or mirtazapine) were added as covariates with treatment group in separate linear mixed models with CMAI score as the dependent variable, there was a significant treatment difference in CMAI score favoring nabilone (atypical antipsychotics: b = −5.75, [−9.66

FIGURE 2. Treatment differences in agitation scores. The middle bar in the boxes represents the median; circles in the boxes represent the mean; the lower and upper ends of the boxes represent the first and third quartiles, respectively. The whiskers represent values within 1.5 times the interquartile range from the upper and lower quartile. *denotes significant treatment differences at a specific visit (Week 2: nabilone: 62.5 ± 19.2 versus placebo 68.3 ± 16.3, (t(32) = −2.39, p = 0.03); Week 6: nabilone: 55.8 ± 15.9 versus placebo: 65.9 ± 13.7, (t(32) = −3.77, p = 0.001). X-axis reflects treatment week regardless of phase.



to −1.83], t(27.5) = −3.0, p = 0.006; antidepressants: b = −4.44 [−7.17 to −1.72], t(29.8) = −3.3, p = 0.002; SSRIs: b = −4.70, [−8.56 to −0.85], t(27.2) = −2.5, p = 0.02; other antidepressants (trazodone or mirtazapine): b = −4.72, [−8.03 to −1.41], t(30.2) = −2.9, p = 0.007). In each of the models, neither psychotropic use nor the treatment x psychotropic use interaction term significantly contributed to the model. An analysis with SNRIs was not completed as only two participants were taking this medication.

DISCUSSION

In this trial, nabilone treatment was associated with a significant reduction in agitation over 6 weeks. This change was greater than that in similar RCTs conducted with atypical antipsychotics[6,29] and antidepressants[30–32] which have reported improvements in CMAI scores of −2.38 points (Cohen's d = 0.2) or less in drug versus placebo. Nabilone was also associated with significantly greater improvements on overall NPS and caregiver burden compared to the aforementioned atypical antipsychotics and antidepressants, except citalopram.

In contrast to other RCTs,[29,30,32,33] which have demonstrated significant cognitive worsening compared to placebo, nabilone was associated with a small improvement on the sMMSE, though this was likely not clinically relevant. In contrast, SIB scores were significantly lower during nabilone compared to placebo phases in the subgroup of patients who were assessed with that instrument. This may be the effect of time, with studies suggesting that severe AD patients could be expected to decline 4.8−5.3 points on the SIB over 6 weeks.[34–36] However, the effect of time cannot account for the drug-placebo differences. An expanded trial is needed to investigate these potential cognitive side effects.

One must also take into account that the SIB was administered in a severe AD population who also had agitation. The length of the SIB, compounded by the population's attention span, meant that the SIB often had to be administered in sections, and

FIGURE 3. Percent of patients per CGIC rating in nabilone and placebo phases.

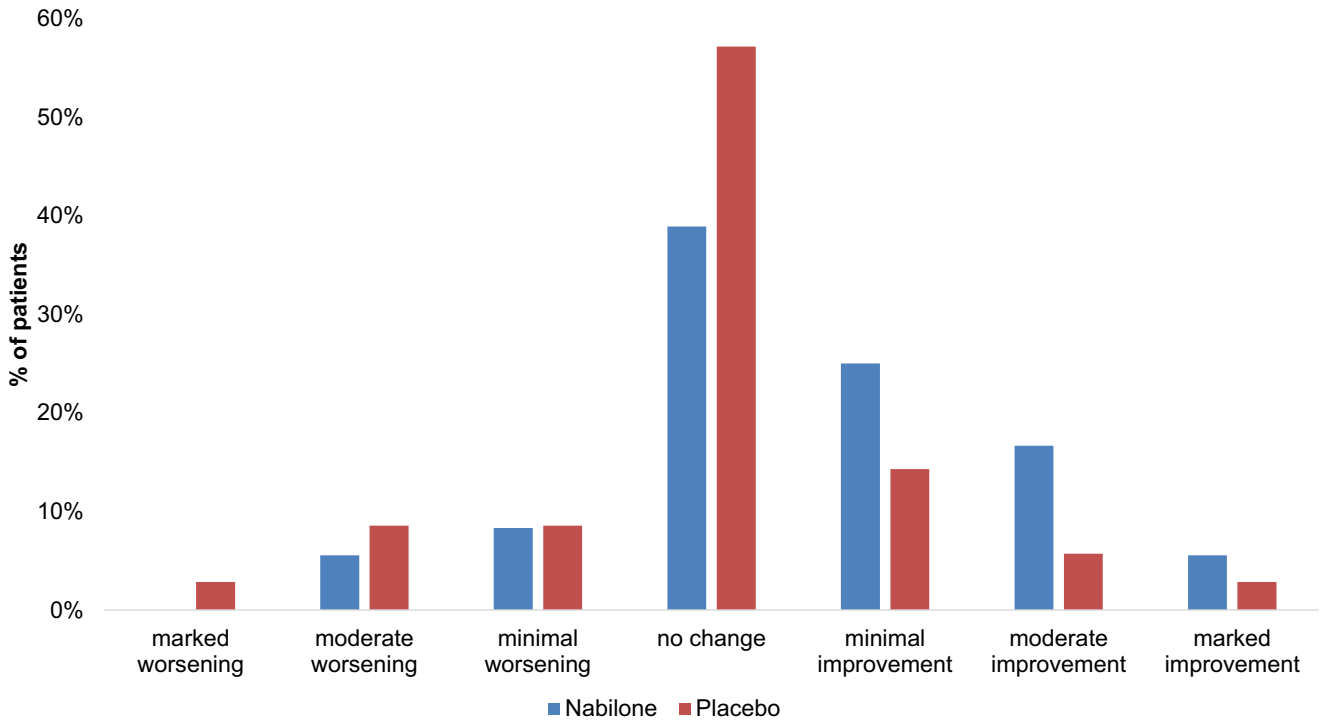


TABLE 2. Incidence of Treatment-Emergent Adverse Events and Severe Adverse Events

	Incidence (N) in Nabilone	Incidence (N) in Placebo
Treatment-Emergent Adverse Events		
Total	31	14
Sedation (including lethargy)	17	6
Treatment limiting sedation	5	1
Falls	8	7
Bradycardia	1	0
Myoclonic Jerk	1	0
Elevated urea levels	1	0
Rash	1	0
Significant increase in NPS	1	2
Dizziness	1	0
Shakiness	0	1
Severe Adverse Events		
Total	5	4
Lethargy	2	0
Death	1	1
Critically high INR	1	0
Myocardial infarction	1	0
Cancer diagnosis	0	1
Pneumothorax	0	1
Sepsis due to UTI	0	1

NPS: neuropsychiatric symptoms; INR: International Normalized Ratio: UTI: urinary tract infection.

therefore SIB scores may not accurately reflect an agitated patient's cognitive status. The short form of the SIB (SIB-S) may rectify these issues. The SIB-S can be administered in 10−15 minutes, and scores are significantly and highly correlated to the original SIB scores (rho = 0.99).[37] Future studies may also consider assessing psychosocial functioning through validated assessments such as the Alzheimer's Disease Cooperative Study Activities of Daily Living Inventory for Severe Alzheimer's Disease (ADCS-ADL-Severe), as cognitive and functional impairment have been shown to correlate with one another in the advanced stages of AD.[38]

Of importance, since we enrolled participants with treatment-resistant agitation, a number of participants were receiving psychotropics. This may be considered a limitation as these medications may exacerbate the treatment differences observed in this study. However, when antipsychotic use, antidepressant use, SSRI use, and other antidepressant use were added as covariates in separate linear mixed models, treatment differences in CMAI total score remained significant, favoring

nabilone, and neither psychotropic use, nor the treatment x psychotropic use interaction term significantly contributed to the model.

Though the efficacy findings are promising, the safety of nabilone is an important consideration as current pharmacologic agents for agitation have high-risk side effect profiles[6] and there is a lack of safer alternatives. We did not identify any significant differences in the number of reported SAEs between treatment groups. Sedation was the most frequently reported TEAE in our study, but was not reported in other trials with cannabinoids in AD.[12,13,24,39,40] This may be attributed to differences in study population as our patients were older, more cognitively impaired, and included a greater proportion of institutionalized patients compared to previous trials with cannabinoids. In addition, previous trials with cannabinoids failed to demonstrate efficacy, and doses may not have been comparable. As our trial included flexible dosing based on tolerability, we were able to investigate treatment-limiting sedation and reported no significant differences between nabilone (13%) and placebo (3%). Furthermore, the incidence of treatment-limiting sedation with nabilone was at the lower limit of the range of incidence rates of previous RCTs, where 8%−28% of patients experienced sedation with treatment compared to 1%−10% with placebo.[6] Those findings suggest that nabilone may be a safer alternative to atypical antipsychotics, and the occurrence of sedation can be reduced through personalized dosing and close monitoring of AEs.

In this RCT, nabilone demonstrated greater improvements in agitation and NPS compared to other cannabinoids such as THC and dronabinol (synthetic THC).[12,13,24,39,40] This may be due to differences in pharmacokinetic and pharmacodynamic properties. THC has not shown efficacy in treating NPS in AD, but has a shorter time to maximum concentration, and a faster absorption period. Dronabinol is a full agonist at CB1/2 receptors, while nabilone is a partial agonist.[41] Dronabinol has a faster onset of action (approximately 30 minutes), while nabilone has a longer duration of action (approximately 8−12 hours) and better bioavailability.[42] Additionally, dronabinol has a half maximal effective concentration (EC_{50}) of 0.017 μM at CB1 and 0.0015 μM at CB2. Nabilone has a lower EC_{50} at CB1 (0.004 μM) and a greater EC_{50} at CB2 (0.016 μM). Therefore, compared to dronabinol, nabilone is four times more potent at CB1, which is consistent with the recommended dose ranges in adults (nabilone: 2−6 mg/day; dronabinol: 5−20 mg/day). While dose equivalency has not been established, nabilone 2−8 mg and dronabinol 10−20 mg have similar subjective, cognitive, and cardiovascular effects in adults.[42] Both nabilone and dronabinol are approved by Health Canada and the Food and Drug Administration in the United States. However, nabilone has been withdrawn from the American market and dronabinol from the Canadian market by their respective manufacturers for unknown reasons.

Agonism at CB1 has been associated with increased feeding, food craving, food enjoyment, and energy deposition of fat into adipose tissues. Clinical studies have also demonstrated that cannabinoids such as THC[43] and dronabinol[44,45] improve nutritional status and weight gain in patients with anorexia. We therefore assessed whether nabilone was associated with increases in nutritional status, weight, and BMI.[46] In AD, weight loss is common and associated with rapid cognitive decline.[47] As such, an intervention that improves nutritional intake and increases BMI is of importance. Though our study did not identify significant improvements in BMI during nabilone compared to placebo, nabilone was associated with significant improvements on the MNA-SF, suggesting potential benefits on nutritional status. Studies reporting improvements in weight and/or BMI have had trial durations of 3 months or greater,[48,49] suggesting that our study may have been too short to notice any significant benefits. The patients enrolled in our study also had a greater BMI at baseline compared to patients enrolled in previous nutrition intervention trials with cannabinoids,[24] which was aimed at patients with anorexia. Our findings may also be confounded as 70% of our enrolled patients resided in a long-term care facility, and may have had less opportunity to increase food intake due to structured meal times, despite having an increase in appetite.

Pain in AD has not only been associated with reduced quality of life,[50] function,[51] and poorer nutritional status,[52] but also increased NPS.[53] Current pharmacologic treatments for pain have been associated with limited to modest efficacy and an increased risk of side effects in those with AD.[54] Preclinical and clinical evidence suggests that targeting the endocannabinoid system may offer benefits for pain through inflammatory processes.[55] Our study did not identify any significant improvements on pain with nabilone. This may be due to our lower average level of pain,

with mean PAINAD scores of 2 compared to 6.7 in a previous study in patients with AD that reported a significant change.[56]

Study strengths included randomization, double-blinding, use of an identical-appearing placebo, and the inclusion of a sufficient washout period between treatment phases. Additionally, this study included the careful assessment of agitation, other NPS, cognition, nutritional status, and pain using scales appropriate to the study population. Study limitations included potential sampling bias as patients recruited were predominantly male; the proportion of males to females was greater in our trial compared to other trials in patients with agitation/aggression.[6,8,29,30] As the study physician, who also assessed the occurrence of AEs, also completed the CGIC, there was the potential for unblinding. However, AEs such as sedation are not only a side effect of nabilone, but also of other medications and medical factors present in AD patients with agitation. Our trial also permitted stable use of concomitant psychotropics. Though the use of these medications could have masked the effects of nabilone, the majority of patients were prescribed one to two psychotropics in this study, reflecting real world settings. Finally, this study had a relatively small sample size, which should be taken into consideration when interpreting results. However, based on our post hoc power analysis, our study was adequately powered.

CONCLUSION

This double-blind crossover RCT is the first to investigate the efficacy and safety of a synthetic cannabinoid in patients with AD. Cannabinoids, such as nabilone, have a distinct pharmacologic profile that may offer an alternative mechanism through which to treat agitation compared to previous pharmacologic therapies with modest efficacy and significant safety concerns. Our findings suggest that nabilone may be an effective treatment for agitation, with potential benefits on overall NPS, caregiver burden, and nutritional status. Side effects such as sedation, and possibly cognitive decline, should be closely monitored to ensure patient safety. This trial supports a larger and longer trial with nabilone to provide confirmatory evidence in this difficult-to-treat patient group.

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